Inverse Transform Method

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Inverse Transform Method

- We define the Probability Integral Transformation

Probability Integral Transformation

Let X have a continuous cdf $F_X(x)$ and define the random variable Y as $Y = F_X(X)$. Then Y is uniformly distributed on $(0,1)$ that is $P(Y \leq y) = y, 0 < y < 1$

Proof

$$
\begin{aligned}
P\{Y \leq y\} &= P\{F(X) \leq y\} \\
&= P\{F^{-1}(F(X)) \leq F^{-1}(y)\} \\
&= P\{X \leq F^{-1}(y)\} \\
&= F(F^{-1}(y)) \\
&= y
\end{aligned}
$$

Steps in deriving random numbers using integral transformation method

1. Derive the cumulative distribution function of $f_X(x)$
2. Derive the inverse function $F_X^{-1}(x)$
3. Write a function to generate random numbers
    i. Generate u from Uniform(0,1)
    ii. compute $x = F_X^{-1}(u)$

Example 1

Write a function to generate n random numbers from the distribution with density $f_X(x) = 3x^2, 0 < x < 1$

- Find the cumulative distribution function of $f_X(x)$ as

$$F_X(x) = x^3, 0 < x < 1$$

- Next we need to compute $F_X^{-1}(u)$ which is

$$F_X^{-1}(u) = u^{\frac{1}{3}}$$

R Function

```r
generate_it <- function(n){
# Generate random numbers
u <- runif(n)
xgen <- u^(1/3)
xgen
}
set.seed(2020)
generate_it(10)
```

Exponential Distribution

The pdf of exponential distribution is $f(x) = \lambda e^{-\lambda x}, x \geq 0$

The cdf of the exponential distribution is

$$F(x) = 1 - e^{-\lambda x}$$

Setting $u = F(x)$ and solving for x we have:

$$u = 1 - e^{-\lambda x}$$

$$x = -\frac{1}{\lambda}\ln(1 - u) = F^{-1}(u)$$

R Function

```r
generate_exp <- function(n,lambda){
# Generate random numbers
u <- runif(n)
xgen <- -(1/lambda) * log(1-u)
xgen
}
set.seed(2020)
generate_exp(10,2)
```

Monte Carlo Integration

Introduction

- Monte Carlo integration provides a numerical approach to solving the problem by re-expressing the problem in the form of an expectation.

- The expectation of any function X can be found by:

$$E[h(X)] = \int_X h(x)f(x)dx$$

where $f(x)$ is the probability density function for x.

Introduction

- If we take $X \sim Uniform(a, b)$ where a, b defines is the limits of our integral such that the PDF is $f(x) = \frac{1}{b-a}$ then a Monte Carlo estimator for $E[h(X)]$ is

$$E[h(X)] = (b-a)\frac{1}{N}\sum_{i=1}^{N} h(X_i)$$

Example 1

Find the Monte Carlo Integral of

$$f(x) = \int_0^1 x^2 dx$$

```r
# Number of random points
n <- 10000

# Generate random points in the interval [0, 1]
x <- runif(n, min = 0, max = 1)

# Function to integrate: f(x) = x^2
f <- function(x) {
  return(x^2)
}

# Evaluate f(x) at random points
f_x <- f(x)

# Monte Carlo estimate of the integral
integral_estimate <- mean(f_x)
integral_estimate
```

Example 2

Find the Monte Carlo Integral of

$$f(x) = \int_0^{\pi} \sin x dx$$

```r
n <- 10000

# Generate random points in the interval [0, pi]
x <- runif(n, min = 0, max = pi)

# Function to integrate: f(x) = sin(x)
f <- function(x) {
  return(sin(x))
}

# Evaluate f(x) at random points
f_x <- f(x)

# Monte Carlo estimate of the integral
integral_estimate <- (pi - 0) * mean(f_x)
integral_estimate
```